IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MATHEMATICS FOR ENGINEERS IIB

Thursday, 9 January 2025, 10:00 AM

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Vlaski
Second Marker(s) : W.T. Pike

© Imperial College London

ELEC50011 PART B: PROBABILITY AND STATISTICS

1. Consider two continuous random variables X and Y that have the joint probability density function (pdf) given in the following equation:

$$f_{X,Y}(x,y) = \begin{cases} ae^{x+y}, & \text{if } x > 0, y > 0 \text{ and } 0 < x+y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Here, $a > 0$ is a constant.

a) Find the value for a that ensures that $f_{X,Y}(x,y)$ describes a valid pdf. [4]

b) Are X and Y independent? Justify your answer. [3]

c) Find the marginal $f_X(x)$. [5]

d) Compute the expected values of X and Y, i.e., $E(X)$ and $E(Y)$. [3]

e) Compute the variance of X, i.e., $Var(X)$. [3]

f) Find the covariance between X and Y, i.e., $Cov(X,Y)$. [4]

g) Find the conditional probability density function of Y given $X = x$. [2]

h) Compute the conditional expectation of Y given $X = x$. [3]

i) Evaluate $E(E(Y|X))$ and compare to $E(X)$ and $E(Y)$. How and why do these relate? [3]

2. You are presented with a bag containing red balls and blue balls. The total number of balls is 100. Of the 100 balls, a fraction of p are red, where $0 < p < 1$ is unkown to you. This means that if you pick a random ball from the bag, the probability that it will be red is p. You would like to estimate p, and with this goal in mind you construct an experiment. You pick balls from the bag with replacement until you see a red ball for the first time, and record in the random variable X the number of tries it took.

a) Formulate an expression for the probability mass function of X and show that it is a valid pmf. What is the name of this distribution? [5]

b) You repeat the experiment n times under identical conditions, collecting the outcome of the i-th experiment in X_i, resulting in an i.i.d. sample of size n, denoted by $X_1, X_2, \ldots, X_n$. Formulate the maximum likelihood estimate of p given $x_1, x_2, ..., x_n$ and show that it maximizes the likelihood. [6]

c) Find the method-of-moments estimate of p based on $x_1, x_2, \ldots, x_n$. [4]

d) Are the resulting estimators unbiased? [5]